Ref. [50] — Loss Landscape Characterization Without Over-Parametrization

Rustem Islamov, Niccolò Ajroldi, Antonio Orvieto, Aurelien Lucchi | NeurIPS 2024

1. The question: can landscape assumptions allow saddles?

The Polyak–Łojasiewicz (PL) condition makes a small gradient imply a small objective gap. That is useful for proving convergence, but it excludes stationary points with suboptimal loss. The authors seek a structural assumption with convergence consequences that can also accommodate saddle points and does not demand an extreme overparameterization argument for each neural network.

2. The actual proposed α–β condition

Write the empirical objective and its component losses as

$$f(x) = \frac{1}{n}\sum_{i=1}^{n} f_i(x), \qquad S = \arg\min f, \qquad x_p \in \mathrm{Proj}(x, S).$$

The central assumption (the paper's Definition 1, Eq. (2)) requires constants $\alpha > \beta > 0$ such that, for every parameter x, a projection x_p can be chosen for which, *for all component losses* i,

$$\boxed{\langle \nabla f_i(x), x - x_p \rangle \geq \alpha\big(f_i(x) - f_i(x_p)\big) - \beta\big(f_i(x) - f_i^*\big).}$$

Here $f_i^* = \inf_x f_i(x)$. The left-hand side measures how much the individual gradient points *away from* a global solution. The first term asks for useful alignment; the second relaxes that requirement. Unlike PL, this condition can be satisfied by some objectives with saddles.

3. How the condition enters the convergence proof

For SGD, $x_{k+1} = x_k - \gamma_k \nabla f_{i_k}(x_k)$, expand the distance to a projected minimizer:

$$\begin{aligned}\|x_{k+1} - x_p\|^2 = \|x_k - x_p\|^2 - 2\gamma_k \langle \nabla f_{i_k}(x_k), x_k - x_p \rangle \\ + \gamma_k^2 \|\nabla f_{i_k}(x_k)\|^2.\end{aligned}$$

Substituting the α–β condition provides a negative term proportional to the function-value gap plus error/noise terms. The paper uses these estimates to establish bounds for SGD, stochastic Polyak stepsizes ($\mathrm{SPS}_{\max}$) and NGN. This display shows the first algebraic step, not the complete proof (which handles projection, sampling and stepsize constraints).

4. Main results: convergence with an error floor

Under the assumptions stated in its theorems, the paper derives for SGD a function-value convergence bound of order $O(K^{-1/2})$ *up to a nonvanishing neighborhood*. Its size includes a term proportional to $\beta\sigma_{\mathrm{int}}^2$, where the interpolation-error quantity measures a mismatch among component minima. The bound is *not* a guarantee that arbitrary finite-width deep networks attain a global minimum exactly.

Paper result. The proposed condition can include saddle-containing landscapes while supporting meaningful gradient-method bounds. The authors test the condition along trajectories of MLPs, CNNs, ResNets and several larger models, including Transformer workloads.

5. Assumptions and key empirical caveat

The convergence theorems require the specified α–β property and additional regularity/stepsize conditions. Empirical tests check the inequality only along *sampled optimization trajectories*, not at every point in $\mathbb{R}^d$. Their experimental findings therefore do not prove that the loss landscape globally satisfies the condition. In **RQ4**, this is a rare attempt to bridge model-specific landscape assumptions and practical optimization, but it is not a replacement for unrestricted saddle-escape theory.

Critical distinction. A bound that approaches a ball around the optimal objective value is not equivalent to exact global convergence. Even checking the condition on many training runs does not establish it over the full parameter space.

6. Our notes

This paper matters because many landscape theorems assume something that rules out saddles or requires unrealistic widths. The α–β condition is more flexible, and the authors actually check it on modern models. What we must emphasize is that the empirical tests are along training paths, and the theory can leave a nonzero error floor, so this does not magically settle global convergence for ordinary neural networks. **Original source:** https://proceedings.neurips.cc/paper_files/paper/2024/file/52f050499cf82fa8efb588e263f6f3a7-Paper-Conference.pdf. **Verification:** Definition 1, Eq. (2), results in Sec. 4, empirical limitations in Sec. 6 checked.